



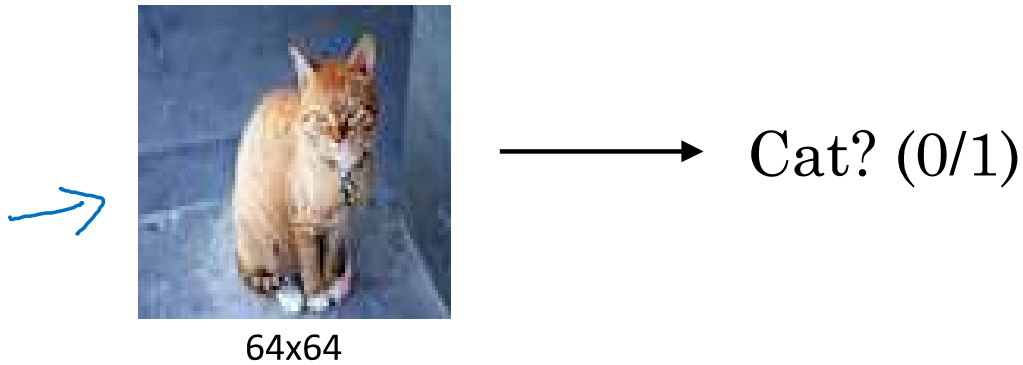
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Convolutional Neural Networks

Computer vision

Computer Vision Problems

Image Classification



Neural Style Transfer

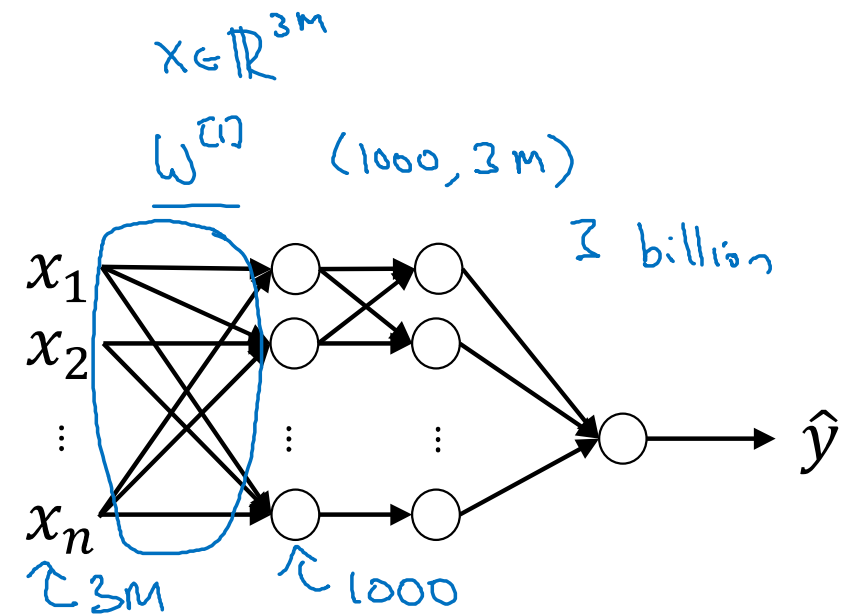
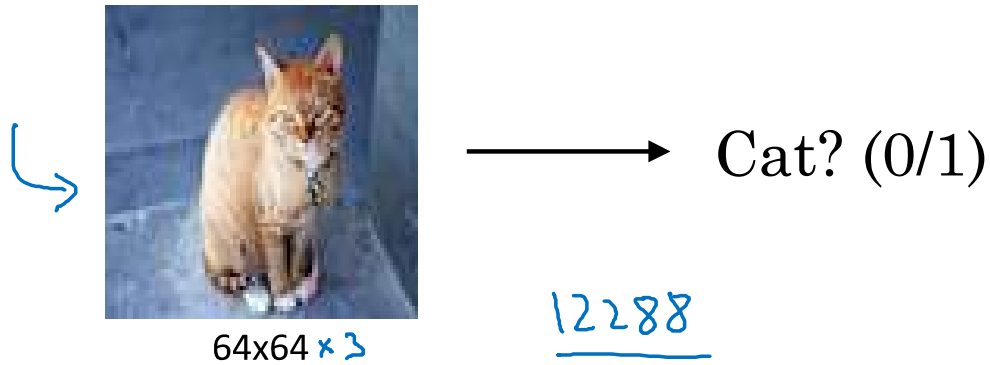


Object detection



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Deep Learning on large images



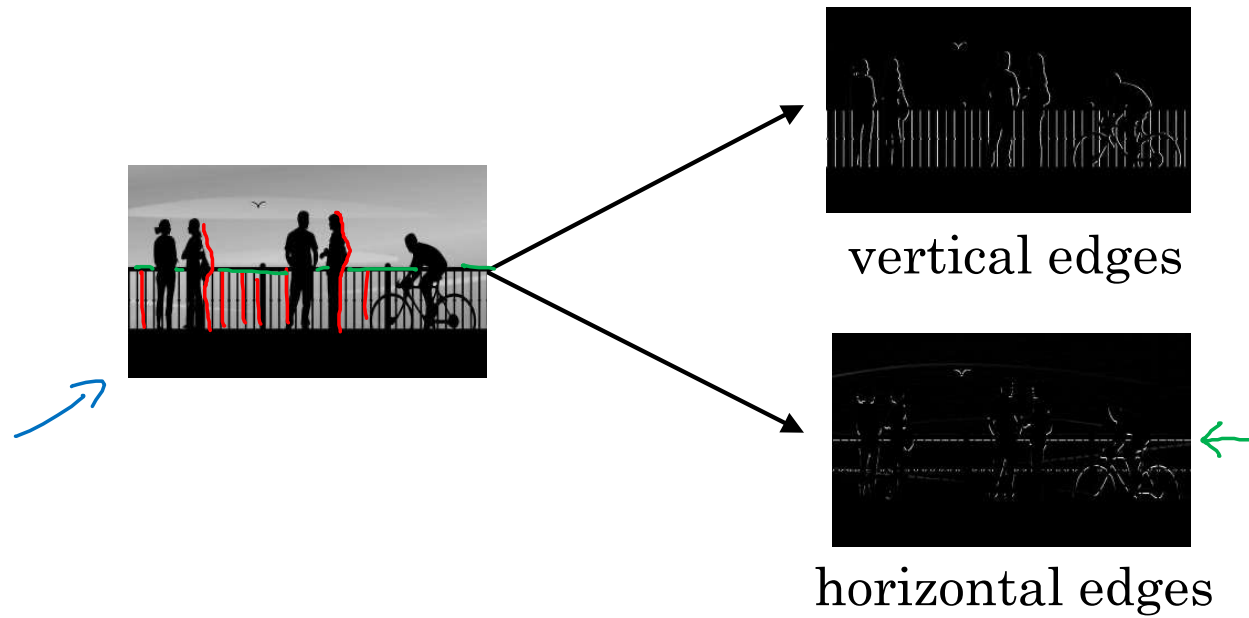
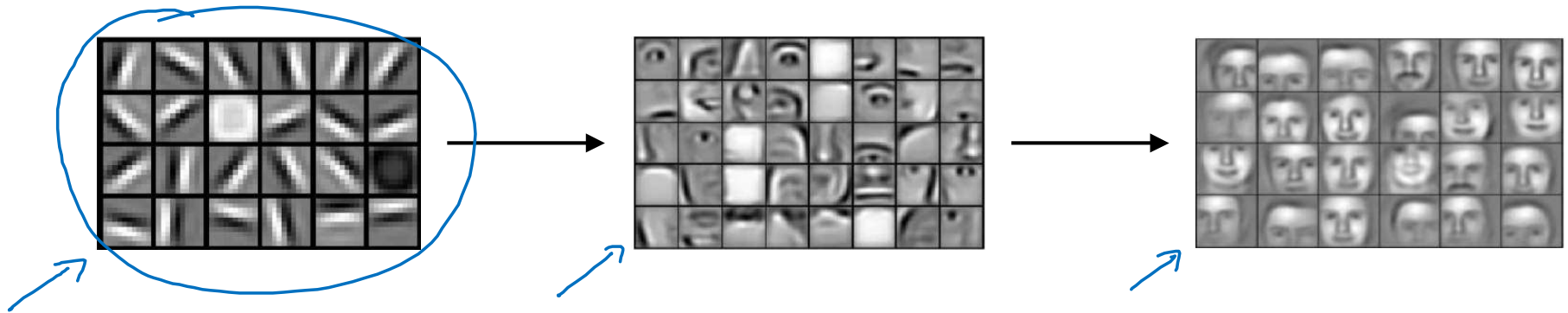


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Convolutional Neural Networks

Edge detection example

Computer Vision Problem

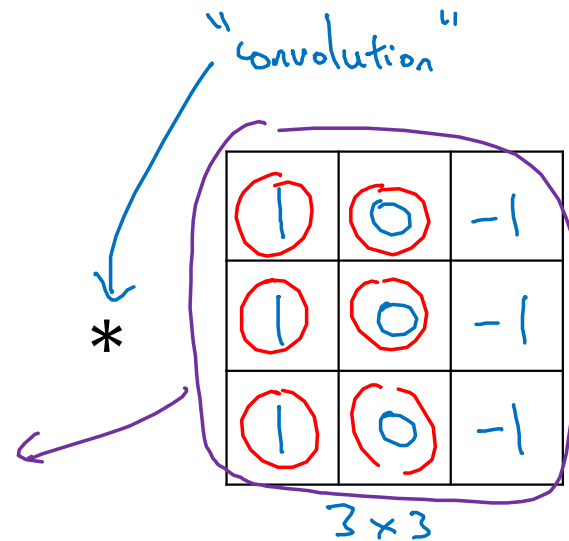


Vertical edge detection

$$\rightarrow 3 \times 1 + 1 \times 1 + 2 \times 1 + 0 \times 0 + 5 \times 0 + 7 \times 0 + 1 \times -1 + 8 \times -1 + 2 \times -1 = -5$$

3	0	1	2	7	4
1	5	8	9	3	1
2	7	2	5	1	3
0	1	3	1	7	8
4	2	1	6	2	8
2	4	5	2	3	9

6x6



=

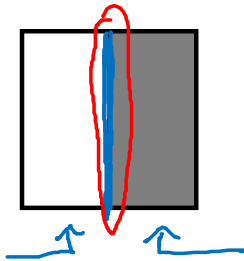
-5	-4	0	8
-10	-2	2	3
0	-2	-4	-7
-3	-2	-3	-16

4x4

Vertical edge detection

10	10	10	0	0	0
10	10	10	0	0	0
10	10	10	0	0	0
10	<u>10</u>	<u>10</u>	<u>0</u>	0	0
10	<u>10</u>	<u>10</u>	<u>0</u>	0	0
10	<u>10</u>	<u>10</u>	<u>0</u>	0	0

6x6

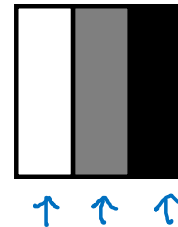


*

1	0	-1
1	0	-1
1	0	-1

3x3

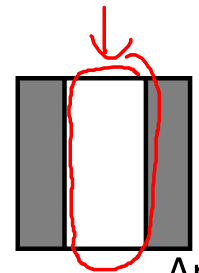
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0	30	30	0
0	30	30	0
0	30	30	0
0	30	30	0

4x4



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Convolutional Neural Networks

More edge
detection

Vertical edge detection examples

10	10	10	0	0	0
10	10	10	0	0	0
10	10	10	0	0	0
10	10	10	0	0	0
10	10	10	0	0	0
10	10	10	0	0	0



0	0	0	10	10	10
0	0	0	10	10	10
0	0	0	10	10	10
0	0	0	10	10	10
0	0	0	10	10	10
0	0	0	10	10	10



*

1	0	-1
1	0	-1
1	0	-1



=

0	30	30	0
0	30	30	0
0	30	30	0
0	30	30	0



*

1	0	-1
1	0	-1
1	0	-1



=

0	-30	-30	0
0	-30	-30	0
0	-30	-30	0
0	-30	-30	0



Vertical and Horizontal Edge Detection



1	0	-1
1	0	-1
1	0	-1

Vertical



1	1	1
0	0	0
-1	-1	-1

Horizontal

10	10	10	0	0	0
10	10	10	0	0	0
10	10	10	0	0	0
0	0	0	10	10	10
0	0	0	10	10	10
0	0	0	10	10	10

6x6

*



1	1	1
0	0	0
-1	-1	-1

=

0	0	0	0
30	10	-10	-30
30	10	-10	-30
0	0	0	0

Learning to detect edges

1	0	-1
1	0	-1
1	0	-1



→

1	0	-1
2	0	-2
1	0	-1

Sobel filter



3	0	-3
10	0	-10
3	0	-3

Scharr filter



3	0	1	2	7	4
1	5	8	9	3	1
2	7	2	5	1	3
0	1	3	1	7	8
4	2	1	6	2	8
2	4	5	2	3	9

convolution
*

W_1	W_2	W_3
W_4	W_5	W_6
W_7	W_8	W_9

3x3

=

45°
70°
73°

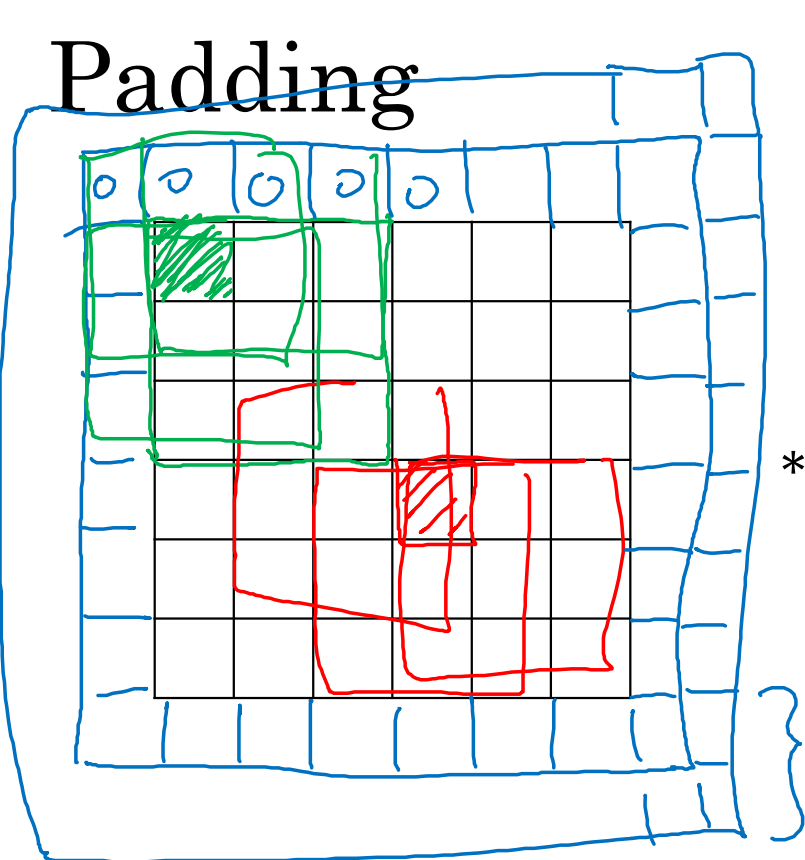


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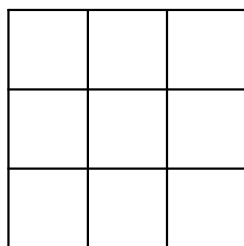
Convolutional Neural Networks

Padding

Padding



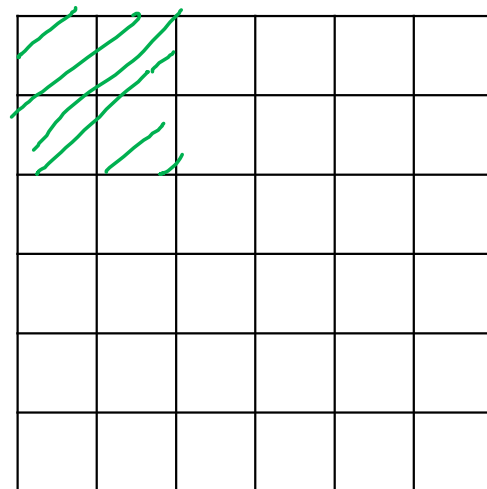
- shrinky output
- throw away info from edge



3x3
f x f

*

=



6x6

6x6 → 8x8
n x n

$n - f + 1 \times n - f + 1$

$6 - 3 + 1 = 4$

p = padding = 1

$n + 2p - f + 1 \times n + 2p - f + 1$

$6 + 2 - 3 + 1 \times \underline{\quad} = 6 \times 6$

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Valid and Same convolutions

→ no padding

“Valid”: $n \times n$ $\times$ $f \times f$ $\rightarrow \frac{n-f+1}{1} \times n-f+1$
 6×6 $\times$ 3×3 $\rightarrow 4 \times 4$

“Same”: Pad so that output size is the same as the input size.

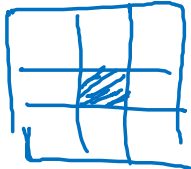
$$n+2p-f+1 \times n+2p-f+1$$

$$\cancel{n+2p-f+1} = \cancel{n} \Rightarrow \boxed{p = \frac{f-1}{2}}$$

3×3 $p = \frac{3-1}{2} = 1$ $\left| \begin{array}{c} 5 \times 5 \\ f=5 \end{array} \right.$

f is usually odd

1x1
3x3
5x5
7x7



$p=2$

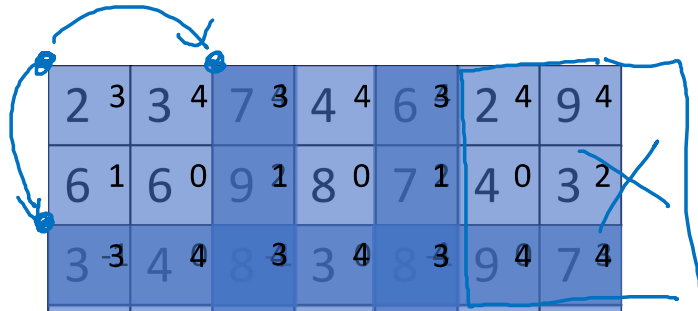


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Convolutional Neural Networks

Strided convolutions

Strided convolution



2	3	3	4	7	3	4	4	6	3	2	4	9	4
6	1	6	0	9	1	8	0	7	1	4	0	3	2
3	3	4	4	8	3	3	4	8	3	9	4	7	4
7	1	8	0	3	1	6	0	6	1	3	0	4	2
4	3	2	4	1	3	8	4	3	3	4	4	6	4
3	1	2	0	4	1	1	0	9	1	8	0	3	2
0	-1	1	0	3	-1	9	0	2	-1	1	0	4	3

7x7

*

3	4	4
1	0	2
-1	0	3

3x3

=



91	100	83
69	91	127
44	72	74

3x3

Stride = 2

$\lfloor z \rfloor = \text{floor}(z)$

$n \times n$ * $f \times f$
 padding p stride s
 $s=2$

$$\left\lfloor \frac{n + 2p - f}{s} + 1 \right\rfloor \times \left\lfloor \frac{n + 2p - f}{s} + 1 \right\rfloor$$

$$\frac{7 + 0 - 3}{2} + 1 = \frac{4}{2} + 1 = 3$$

Summary of convolutions

$n \times n$ image $f \times f$ filter

padding p stride s

Output Size:

$$\left\lfloor \frac{n+2p-f}{s} + 1 \right\rfloor \times \left\lfloor \underbrace{\frac{n+2p-f}{s}} + 1 \right\rfloor$$

Technical note on cross-correlation vs. convolution

Convolution in math textbook:



2 ⁷	3 ²	7 ⁵	4	6	2
6 ⁹	6 ⁰	9 ⁴	8	7	4
3 ¹	4 ¹	8 ³	3	8	9
7	8	3	6	6	3
4	2	1	8	3	4
3	2	4	1	9	8

*

3	4	5
1	0	2
-1	9	7

7	2	5
9	0	4
-1	1	3

=

$$(A * B) * C = A * (B * C)$$

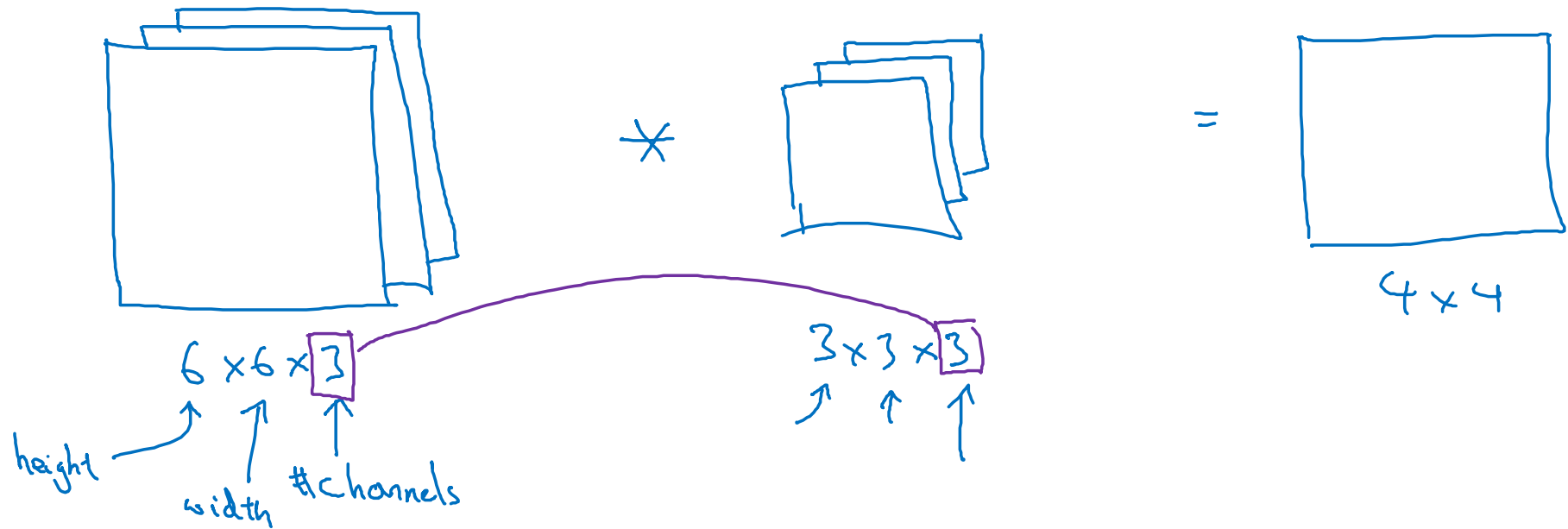


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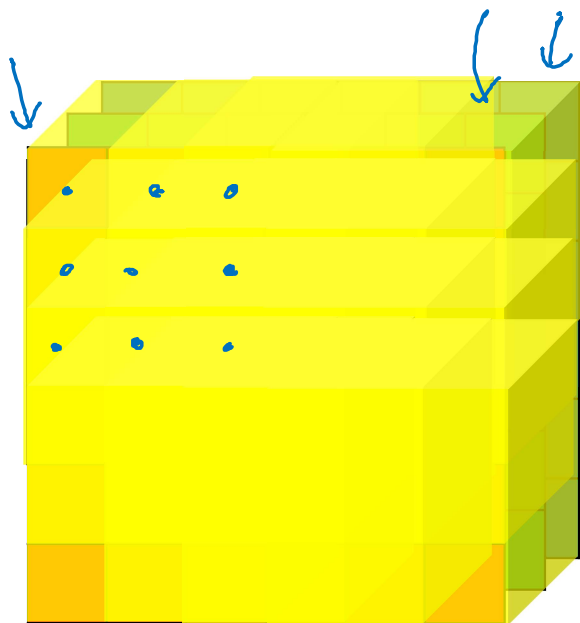
Convolutional Neural Networks

Convolutions over volumes

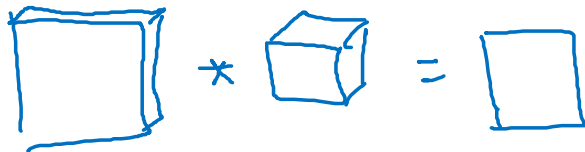
Convolutions on RGB images



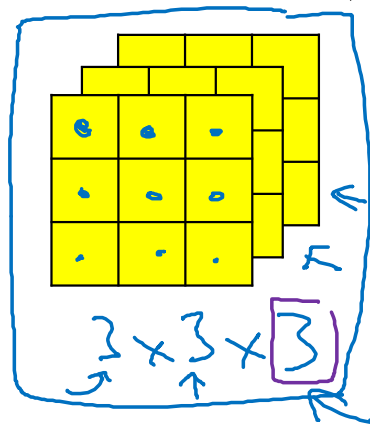
Convolutions on RGB image



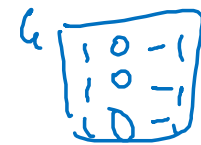
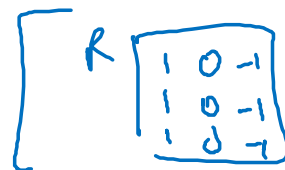
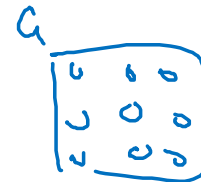
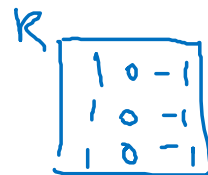
$6 \times 6 \times 3$
 $\uparrow \quad \uparrow \quad \uparrow$



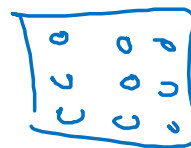
*



27 numbers

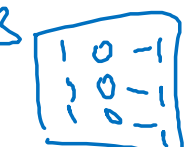


B

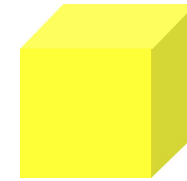


$\rightarrow 3 \times 3 \times 3$

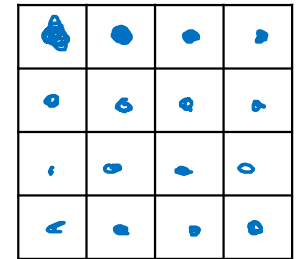
B



$\rightarrow 3 \times 3 \times 3$

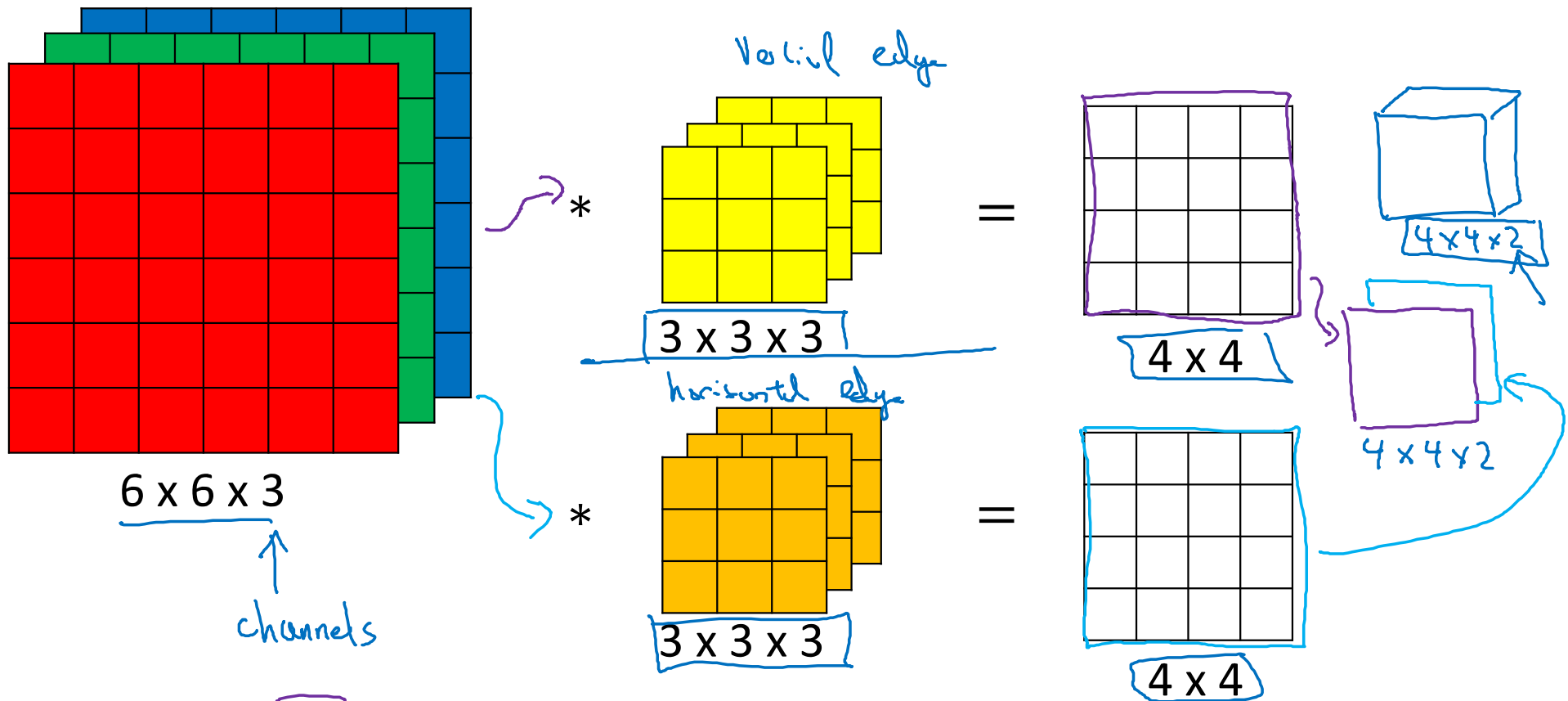


=



4×4

Multiple filters



Summary: $n \times n \times n_c$ $\times$ $f \times f \times n_c$ $\rightarrow$ $\frac{n-f+1}{4} \times \frac{n-f+1}{4} \times n'_c$

$6 \times 6 \times 3$ $3 \times 3 \times 3$ $4 \times 4 \times 2$ $\uparrow$ #filters

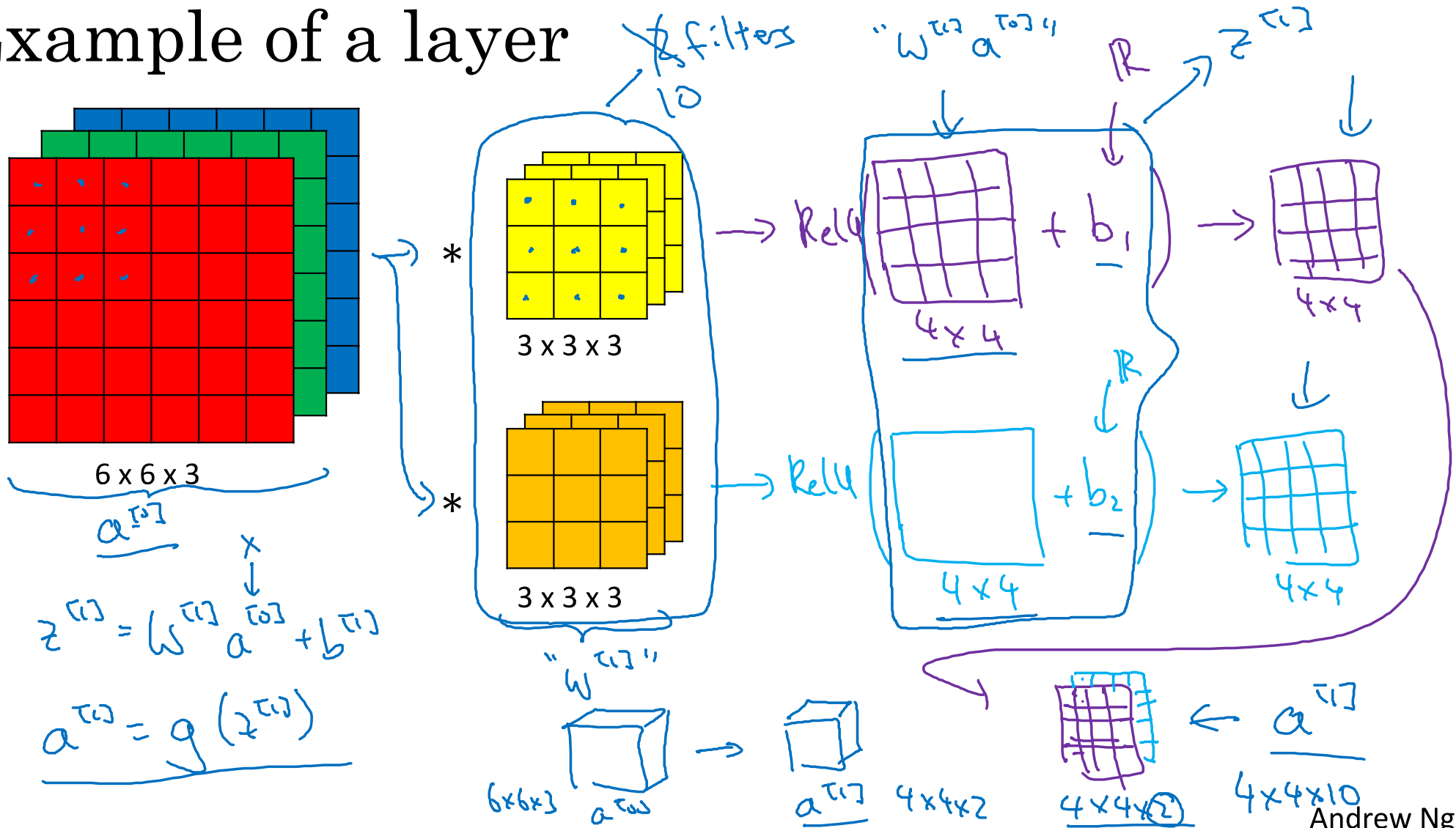


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Convolutional Neural Networks

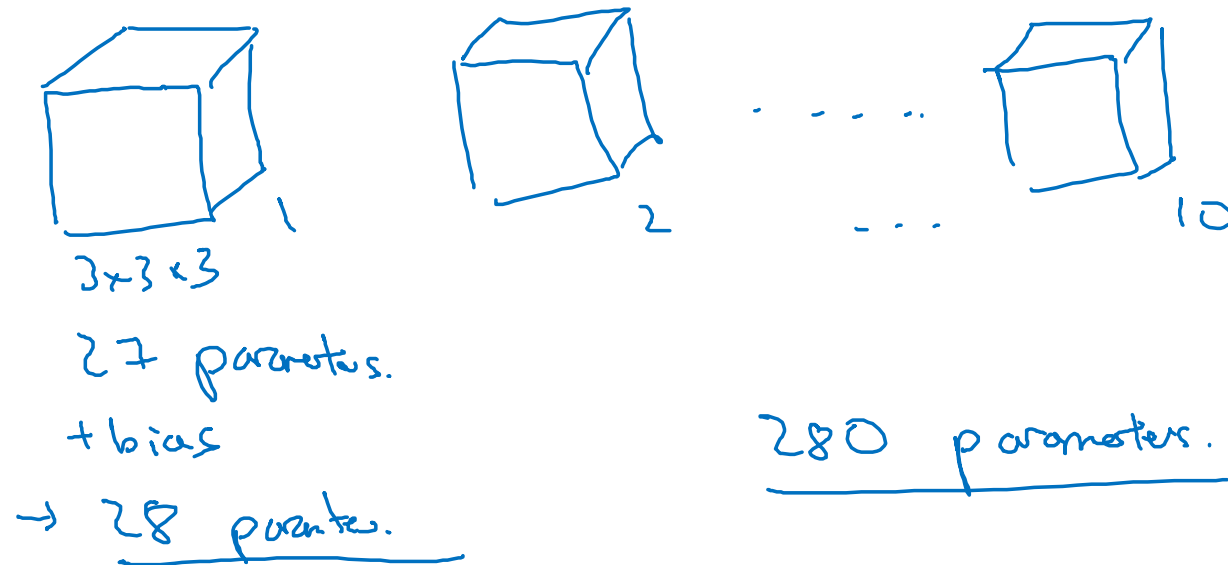
One layer of a
convolutional
network

Example of a layer



Number of parameters in one layer

If you have 10 filters that are $3 \times 3 \times 3$ in one layer of a neural network, how many parameters does that layer have?



Summary of notation

If layer l is a convolution layer:

$f^{[l]}$ = filter size

$p^{[l]}$ = padding

$s^{[l]}$ = stride

$n_c^{[l]}$ = number of filters

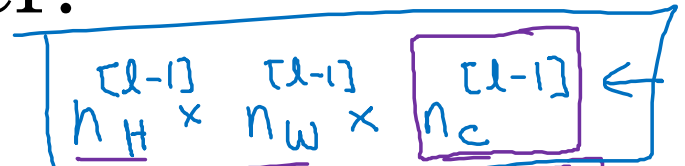
→ Each filter is: $f^{[l]} \times f^{[l]} \times n_c^{[l-1]}$

Activations: $a^{[l]} \rightarrow n_H^{[l]} \times n_W^{[l]} \times n_c^{[l]}$

Weights: $f^{[l]} \times f^{[l]} \times n_c^{[l-1]} \times n_c^{[l]}$

bias: $n_c^{[l]} - (1, 1, 1, n_c^{[l]})$ ← #filters in layer l.

Input:



Output:



$$n_{HW}^{[l]} = \left\lfloor \frac{n_H^{[l-1]} + 2p^{[l]} - f^{[l]}}{s^{[l]}} + 1 \right\rfloor$$

$$A^{[l]} \rightarrow m \times \underbrace{n_H^{[l]} \times n_W^{[l]} \times n_c^{[l]}}_{n_c^{[l]} \times n_H^{[l]} \times n_W^{[l]}}$$

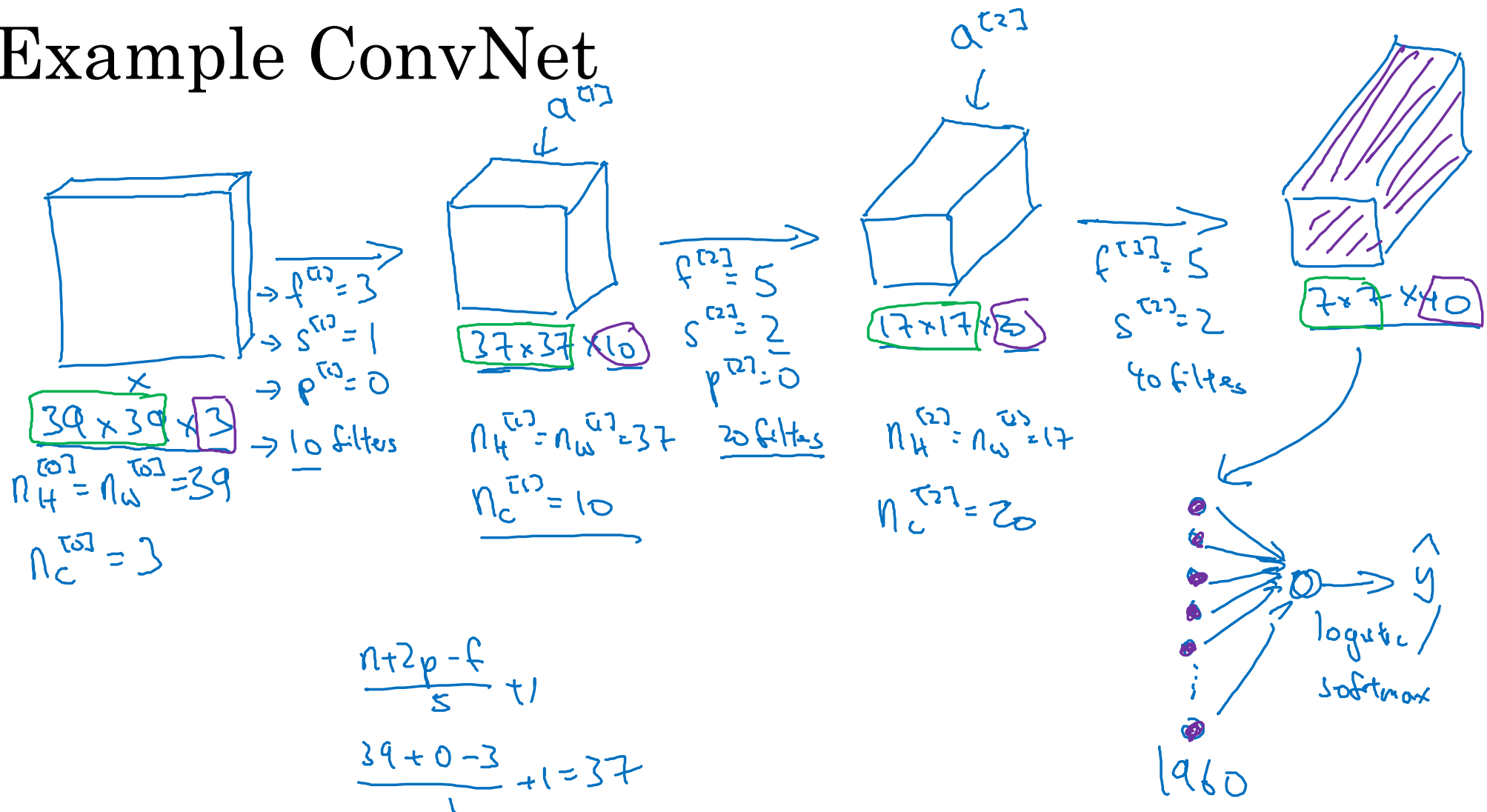


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Convolutional Neural Networks

A simple convolution
network example

Example ConvNet



Types of layer in a convolutional network:

- Convolution (conv) ←
 - Pooling (pool) ←
 - Fully connected (FC) ←
- 



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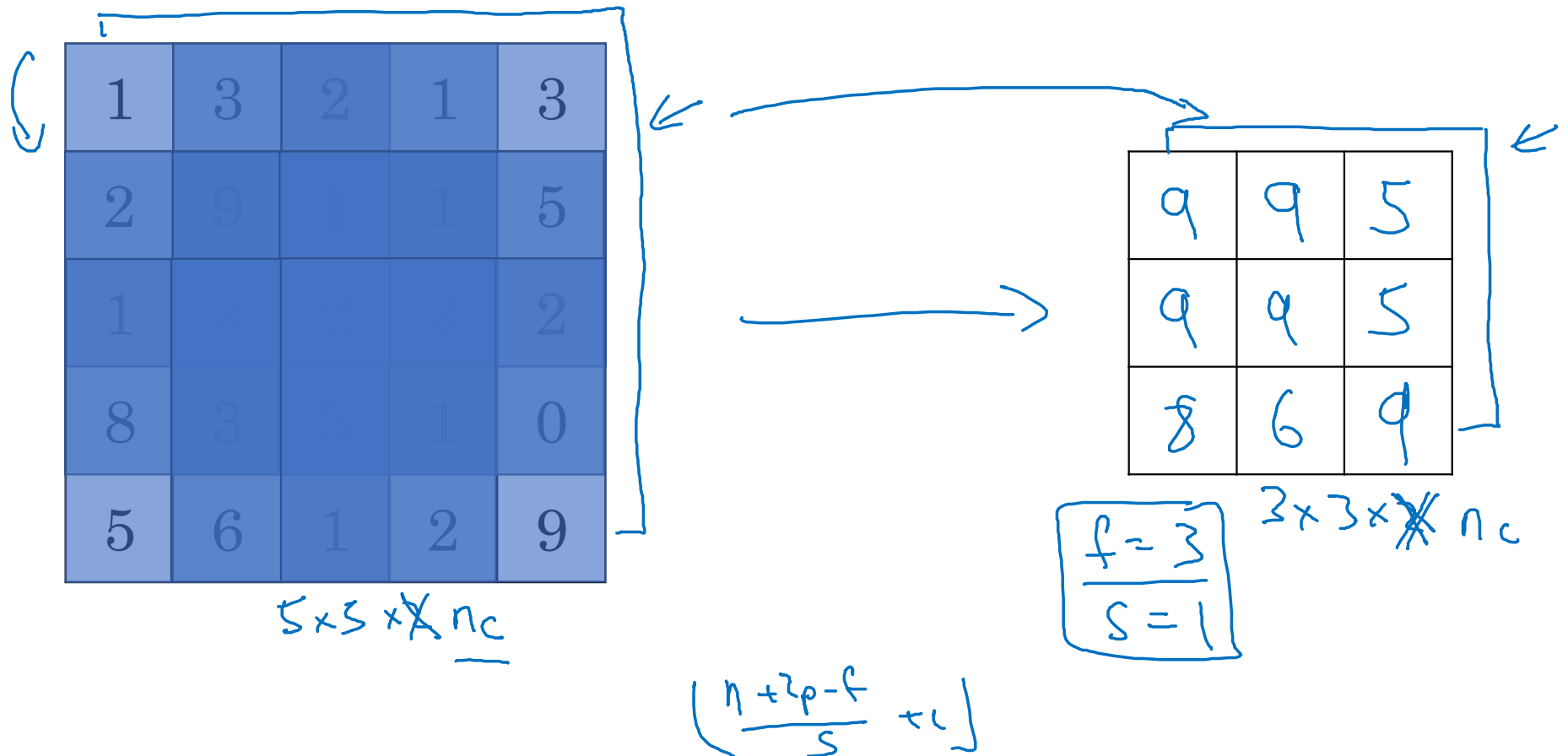
Convolutional Neural Networks

Pooling layers

Pooling layer: Max pooling

1	3	2	1
2	9	1	1
1	3	2	3
5	6	1	2

Pooling layer: Max pooling



Pooling layer: Average pooling

1	3	2	1
2	9	1	1
1	4	2	3
5	6	1	2



3.75	1.25
4	2

$$f=2$$

$$s=2$$

$$\underline{7 \times 7 \times 1000} \rightarrow 1 \times 1 \times 1000$$

Summary of pooling

Hyperparameters:

f : filter size

$$f=2, s=2$$

s : stride

$$f=3, s=2$$

Max or average pooling

~~$\Rightarrow p$: padding.~~

No parameters to learn!

$$N_H \times N_W \times \underline{N_C}$$

$$\downarrow$$
$$\left\lfloor \frac{N_H - f}{s} + 1 \right\rfloor \times \left\lfloor \frac{N_W - f}{s} + 1 \right\rfloor$$
$$\times \underline{N_C}$$

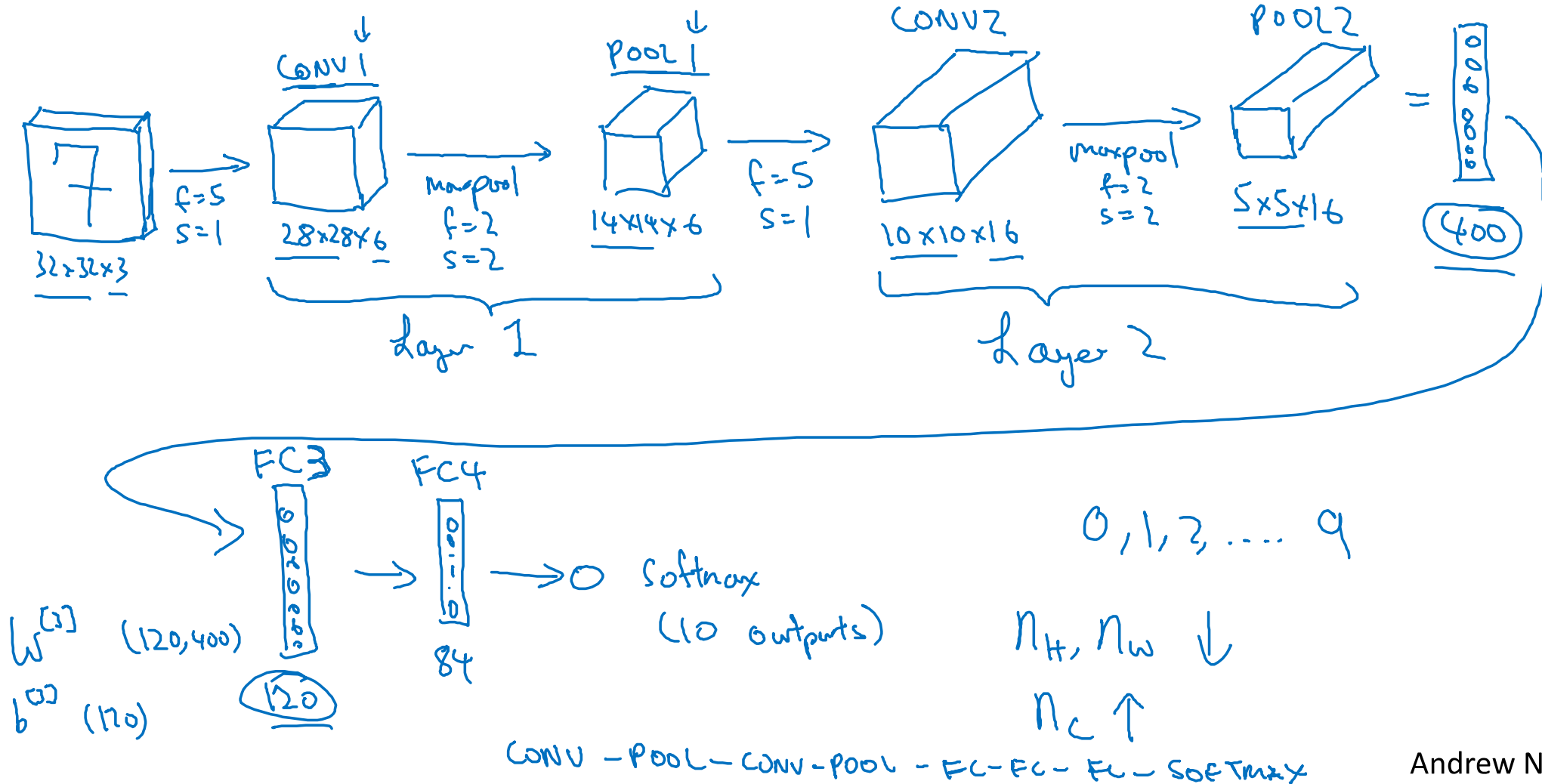


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Convolutional Neural Networks

Convolutional neural network example

Neural network example (LeNet-5)



Neural network example

	Activation shape	Activation Size	# parameters
Input:	(32,32,3)	3,072 $a^{[0]}$	0
CONV1 (f=5, s=1)	(28,28,8)	<u>6,272</u>	608 ←
POOL1	(14,14,8)	<u>1,568</u>	0 ←
CONV2 (f=5, s=1)	(10,10,16)	<u>1,600</u>	3216 ←
POOL2	(5,5,16)	<u>400</u>	0 ←
FC3	(120,1)	<u>120</u>	48120 }
FC4	(84,1)	<u>84</u>	10164 }
Softmax	(10,1)	<u>10</u>	850

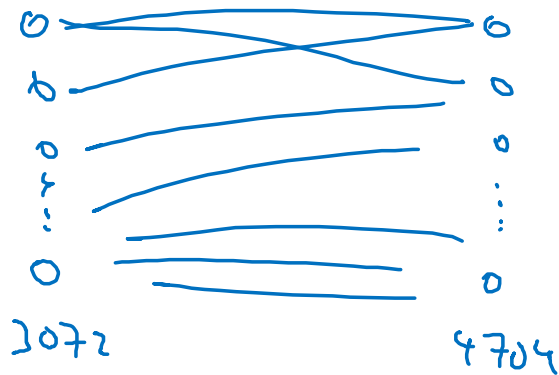
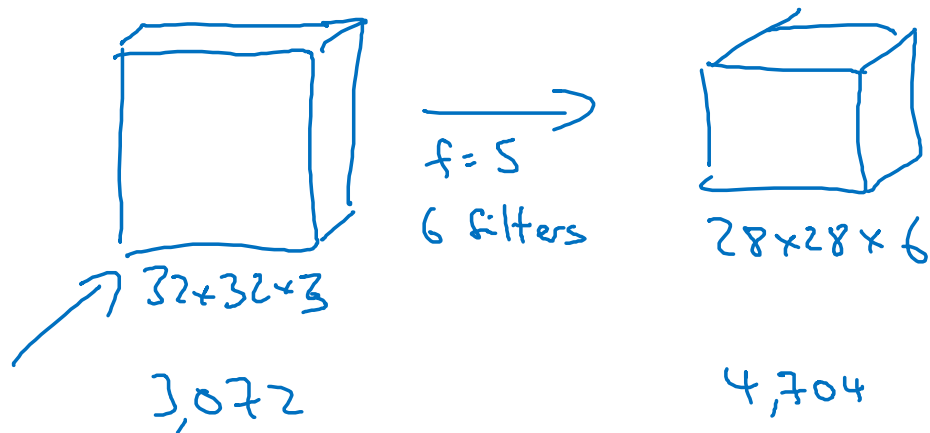


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Convolutional Neural Networks

Why convolutions?

Why convolutions



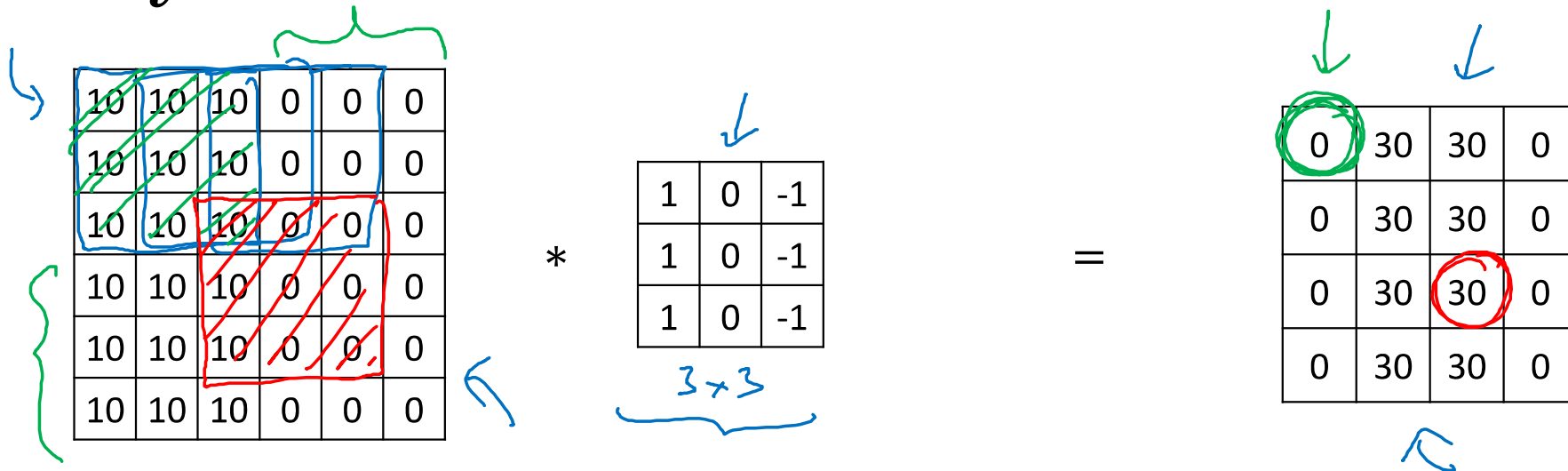
$$5 \times 5 = 25$$

$$+ 1 = 26$$

$$6 \times 26 = 156 \text{ parameters}$$

$$3,072 \times 4,704 \approx \underline{14M}$$

Why convolutions

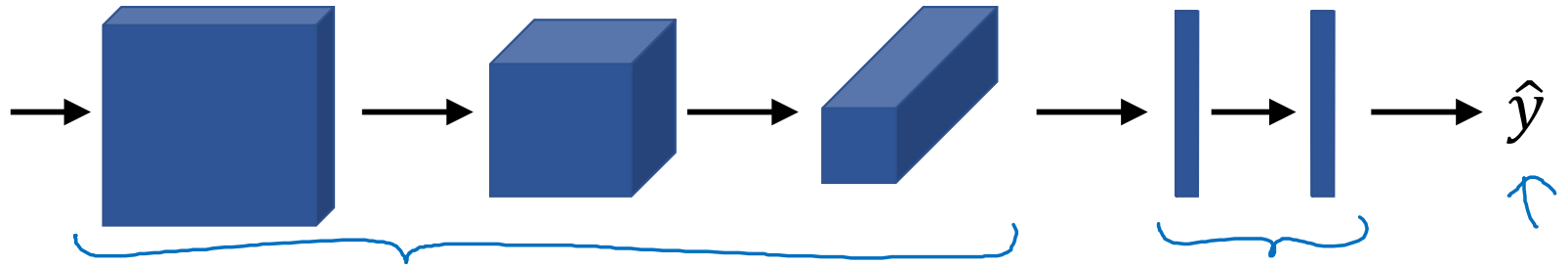


Parameter sharing: A feature detector (such as a vertical edge detector) that's useful in one part of the image is probably useful in another part of the image.

→ **Sparsity of connections:** In each layer, each output value depends only on a small number of inputs.

Putting it together

Training set $(x^{(1)}, y^{(1)}) \dots (x^{(m)}, y^{(m)})$.



$$\text{Cost } J = \frac{1}{m} \sum_{i=1}^m \mathcal{L}(\hat{y}^{(i)}, y^{(i)})$$

Use gradient descent to optimize parameters to reduce J